Chapter 7

Empirical Analysis of Emerging Risk Measures

The previous chapter introduced four complementary measures designed to capture different dimensions of risk emergence in financial news: Novelty, Semantic Drift, News Coverage, and Burst Attention. The present chapter examines how they behave empirically over time and in the five risk themes considered in the study.

Particular attention is paid to the temporal dimension of the measures. Risk emergence is inherently dynamic: a theme may receive sustained attention without generating substantially new information, while another may experience a sudden change in semantic content before becoming widely discussed. Examining the indicators over time therefore allows the analysis to distinguish between changes in informational novelty, semantic content, news attention, and abnormal increases in attention.

Special attention is paid to Cyber Risk.

7.1 Novelty

The temporal dimension is important because changes in Novelty identify periods in which incoming information becomes more or less similar to the recent information associated with that theme.
Figure 7.1 shows monthly mean Novelty separately for Cyber Risk, Climate Risk, Pandemic Risk, Supply Chain Risk, and AI-Related Risk.

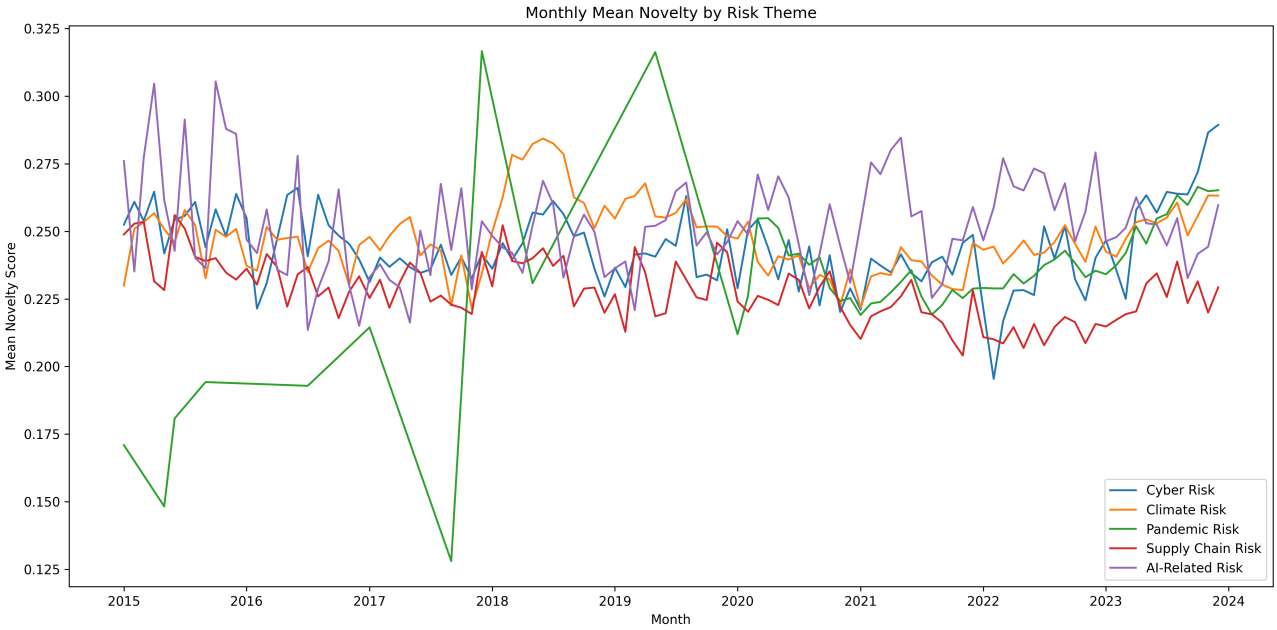


Figure 7.1: Monthly Novelty by Risk Theme, 2015–2023

As expected the five risk themes show quite different patterns.

Climate Risk shows a clear increase in Novelty during 2018. The monthly average rises at the beginning of the year and remains relatively high for several months. After this period, Novelty gradually decreases and reaches lower levels around 2020 and 2021. A moderate increase can then be seen towards the end of the sample. Overall, 2018 stands out as the period in which Climate-related information was particularly novel compared with the rest of its time series.

Supply Chain Risk follows a different path. Novelty is generally higher in the first years of the sample and then slowly decreases. The lowest levels are observed around 2021 and 2022. In 2023, the series starts to increase again, although the change is relatively moderate. There is therefore no single period in which Supply Chain-related information suddenly becomes much more novel. The change is more gradual.

Pandemic Risk needs to be read with more caution. Before 2020, there are very few Pandemic-related articles in the dataset. For this reason, the large movements visible in the first part of the series can be driven by only one or a few observations and should not be treated as strong evidence of a change in Novelty. The series becomes much more informative from 2020, when the number of Pandemic-related articles increases substantially. Novelty decreases after the beginning of this period and reaches lower levels around 2021. It then starts to rise again, with a clearer increase during 2023.

AI-Related Risk is quite volatile, especially in the first years of the sample. Here too, the early values need to be considered together with the relatively small number of AI-related articles available at the time. In the later years the series becomes more stable. Novelty is relatively high in some periods around 2021 and 2022, but there is no clear increase during 2023. This is interesting because the number of AI-related articles grows strongly in that year. More news about a risk, therefore, does not necessarily mean that the information itself is becoming more novel.

Cyber Risk also changes considerably over time. For most of the sample, monthly Novelty moves within a relatively stable range, although there are several short-term increases and decreases. A particularly low value is visible at the beginning of 2022. Novelty then recovers and starts to rise more clearly during 2023. By the end of the year, the monthly average reaches the highest values observed for Cyber Risk in the sample.

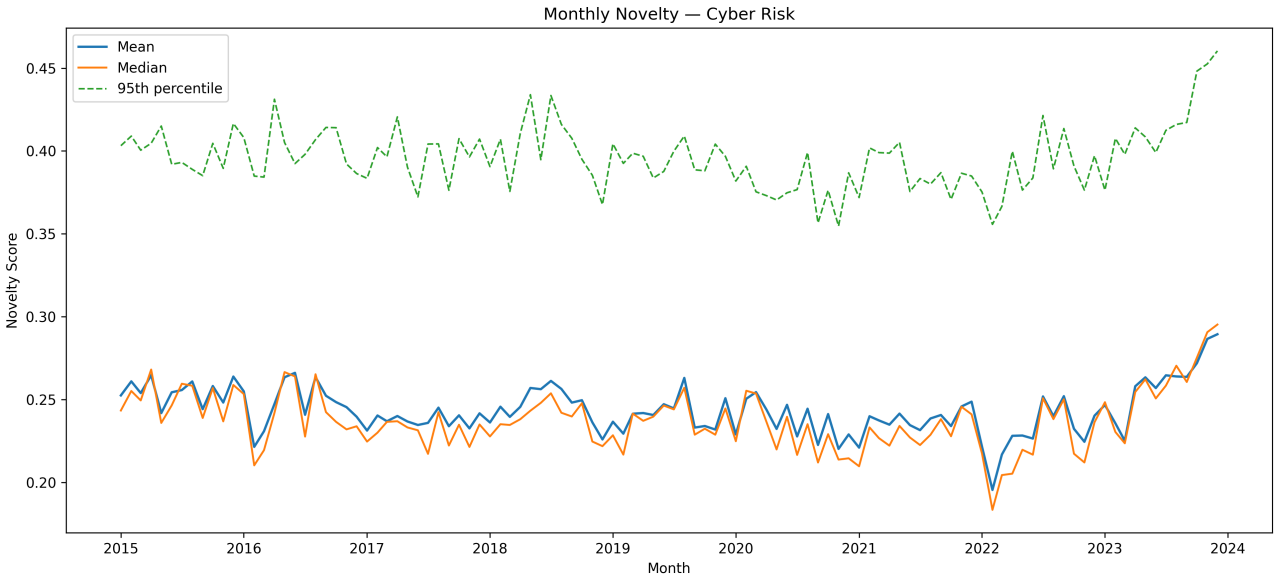


Figure 7.2: Monthly Novelty for Cyber Risk, 2015–2023

For most of the sample, the mean and median remain close to each other. Novelty is somewhat lower during parts of 2020 and 2021, and both measures fall sharply at the beginning of 2022. They then recover during the following months.

The most noticeable change comes in 2023. During the second half of the year, both the mean and the median start to increase. The rise becomes stronger in the final months, when they reach their highest levels in the series. Since the median increases together with the mean, this change does not appear to be driven only by a few highly novel articles. The typical Cyber-related article is also becoming more novel relative to the information that was available before it. The 95th percentile adds another part to the picture. It represents the upper end of the Novelty distribution and therefore captures the articles with particularly high Novelty scores. This measure also rises strongly at the end of 2023 and reaches its highest level in the sample. The increase is therefore visible both in the centre of the distribution and among the most novel Cyber-related articles. Taken together, these results make late 2023 the clearest period of high Novelty for Cyber Risk in the sample. Cyber-related articles become more different from the information that had recently been available, and the change is not limited to a small group of extreme observations. This does not mean that Cyber Risk itself emerged in late 2023. Novelty only captures one part of the process. What we can say from this indicator is that the information related to Cyber Risk became particularly novel during this period. The next indicators will help determine whether this change was also accompanied by changes in the content and attention associated with the risk.

7.2 Semantic Drift

Figure 7.3 shows the monthly Semantic Drift for the five risk themes. Unlike Novelty, not every risk has an observation in every month. As explained in the previous chapter, Semantic Drift is only calculated when there is enough thematic information to construct a reliable monthly representation. Missing periods should therefore not be interpreted as zero drift.

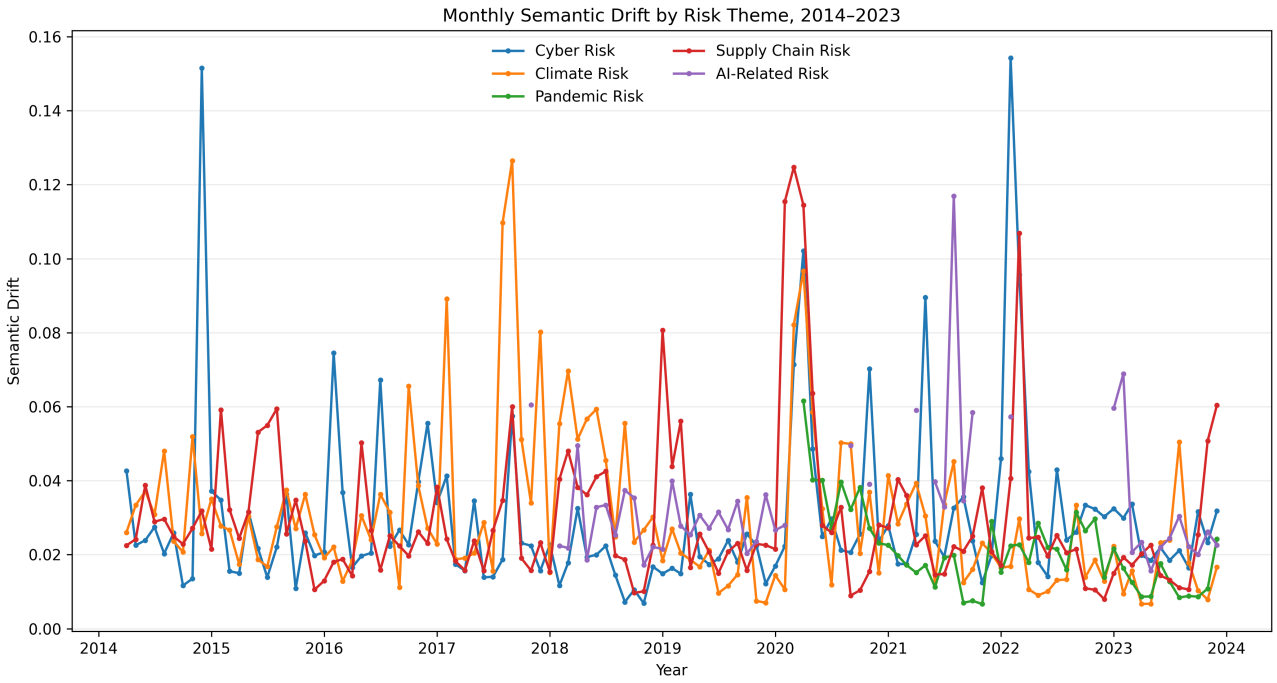


Figure 7.3: Monthly Semantic Drift by Risk Theme, 2014–2023

Most Semantic Drift values remain relatively low over the sample, but the series are interrupted by a number of clear peaks. These peaks are the most interesting part of the indicator. They identify months in which the content associated with a risk changes more strongly relative to its recent history.

Climate Risk shows several periods of higher Semantic Drift, especially between 2016 and 2018. The largest change appears around 2017, when Drift rises above 0.12. Other increases are visible before and after this point, but they are generally smaller. From 2019 onwards, the series is more stable, apart from a noticeable increase around the beginning of 2020. The graph therefore suggests that the content associated with Climate Risk changed more frequently during the earlier part of the sample and became relatively more stable in later years.

Supply Chain Risk also contains several clear changes. One appears around 2019, followed by a much stronger increase at the beginning of 2020. During this period, Semantic Drift rises above 0.12 before falling back towards its usual range. Another large increase is visible in early 2022. These movements suggest that the meaning and content associated with Supply Chain Risk did not change gradually throughout the sample. Instead, the series is mostly stable and is interrupted by specific periods of stronger semantic change.

Pandemic Risk is only available from 2020 onwards because earlier months do not contain enough

thematic information to construct a reliable monthly representation. The first available observations show relatively high Drift, followed by a decline during 2020 and 2021. After this initial period, most values remain at lower levels, generally with only moderate monthly movements. This suggests that the largest semantic adjustment occurred when Pandemic Risk first became strongly represented in the news, while its content became more stable afterwards.

AI-Related Risk is also unavailable for several parts of the sample. Where the indicator can be calculated, most observations remain at relatively moderate levels. The main exception occurs around the second half of 2021, when Semantic Drift rises sharply and reaches a value above 0.11. Other smaller increases appear in 2022 and 2023. The series therefore points to a few specific periods of semantic change rather than a continuous shift in AI-related content.

Cyber Risk shows several isolated peaks across the full sample. Some are visible in 2014, 2016, 2020, and 2021, but the largest occurs at the beginning of 2022.

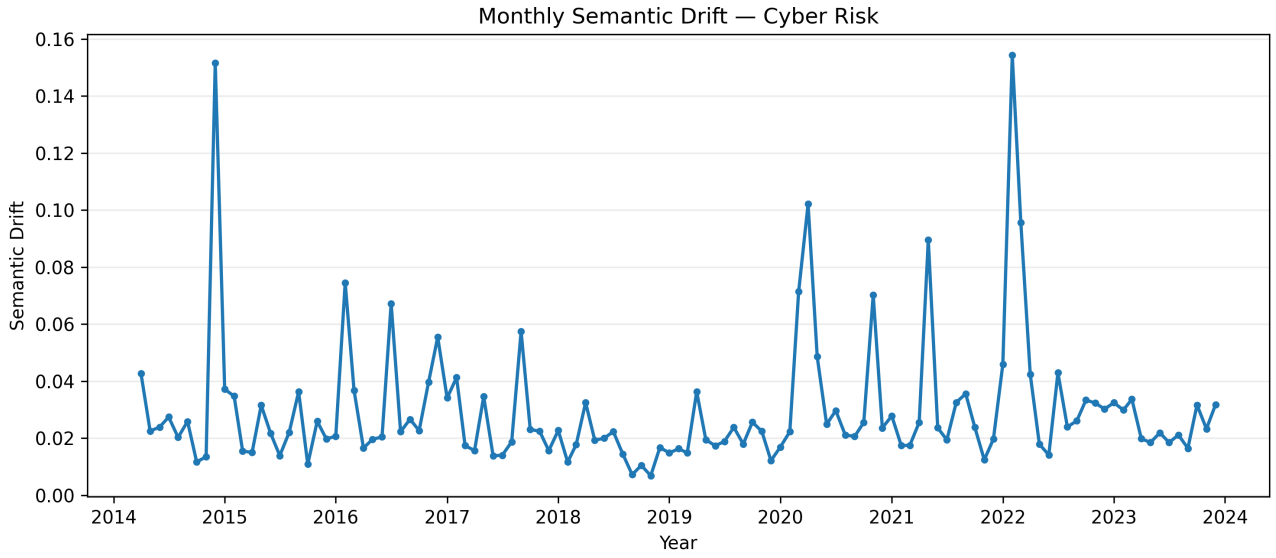


Figure 7.4: Monthly Semantic Drift for Cyber Risk, 2014–2023

For most of the sample, Cyber Semantic Drift remains relatively low. Values are often around 0.02–0.03, suggesting that the monthly semantic representation of Cyber Risk usually remains fairly close to its recent history. This does not mean that the content is unchanged. Rather, most monthly changes are small compared with the much larger movements that appear at specific points in the series.

The first major peak occurs towards the end of 2014, when Semantic Drift rises above 0.15. The increase is sharp and short-lived. Drift returns quickly to much lower levels in the following months. Several smaller peaks appear during 2016 and 2017, but none reaches the same magnitude.

A new period of stronger change appears in 2020. Semantic Drift rises above 0.10 before returning towards its usual range. Another noticeable increase occurs during 2021. These episodes show that the semantic content of Cyber Risk changes at several points in the sample, rather than following a smooth long-term trend.

The largest value in the Cyber series occurs in February 2022, when Semantic Drift reaches approximately 0.154. This is a particularly clear departure from the surrounding months. The increase is followed by another relatively high observation and then by a rapid return towards lower levels. February 2022

therefore represents the strongest semantic change identified for Cyber Risk in the analytical period. The rest of 2022 and 2023 looks quite different. Drift returns to much lower values and remains relatively stable, including during the final months of 2023. This is particularly interesting when compared with the Novelty results. Novelty increases strongly towards the end of 2023, but Semantic Drift does not show a similar increase. The two indicators are therefore capturing different aspects of the Cyber information environment.

The late-2023 Cyber articles become more novel relative to previously available information, but this does not appear to coincide with a major change in the overall semantic representation of Cyber Risk. By contrast, February 2022 is characterised by a strong change in the content associated with the theme. This difference is precisely why Novelty and Semantic Drift are treated as separate dimensions of risk emergence.

7.3 Coverage

Figure 7.5 shows the evolution of News Coverage for the five risk themes between 2014 and 2023. The differences across risks are large, both in terms of their overall level of attention and in the way this attention changes over time.

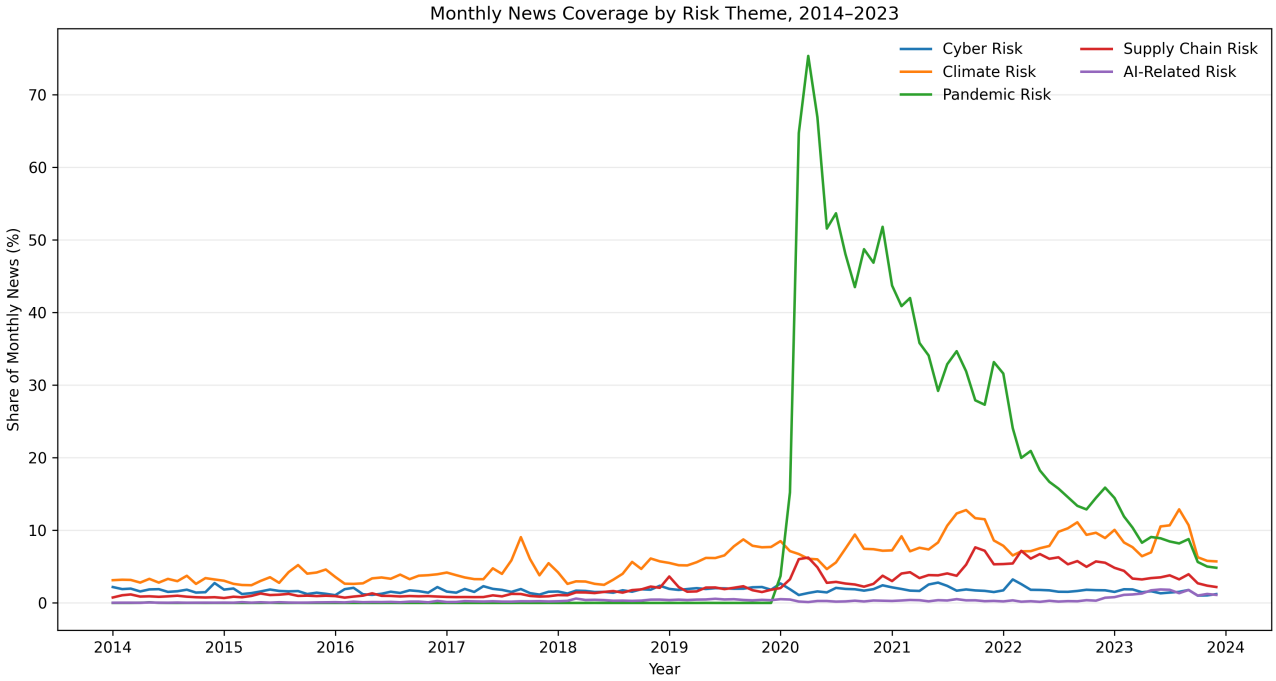


Figure 7.5: Monthly News Coverage by Risk Theme, 2014–2023

Pandemic Risk shows by far the largest change in Coverage. Before 2020, its presence in the news is very limited. This changes abruptly at the beginning of 2020, when Coverage rises to more than 70% of monthly news at its highest point. Attention remains very high during the rest of 2020 and 2021, although it gradually declines. The downward trend continues through 2022 and 2023. By the end of the sample, Pandemic Risk still appears in the news, but its Coverage is much closer to the levels of the other major risk themes.

Climate Risk receives a more persistent level of attention throughout the sample. Coverage is

already relatively high in the early years and generally increases over time. Several periods stand out, including an increase around 2017, another around 2019–2020, and a stronger rise during 2021. Coverage reaches some of its highest levels in 2021 and again in 2023. Unlike Pandemic Risk, the evolution of Climate Risk is therefore not dominated by a single exceptional episode. Its presence in the news grows through several different periods.

Supply Chain Risk is initially less prominent, but its Coverage starts to increase towards the end of the 2010s. A clear rise appears in 2020, followed by another increase during 2021. Coverage remains relatively high through much of 2022 before declining during 2023. The series therefore shows a transition from a relatively limited presence in the earlier years to a much more visible role in the news after 2020.

AI-Related Risk remains the least covered theme for most of the sample. Its share of monthly news is very small during the earlier years and increases only gradually. The clearest change occurs in 2023, when AI Coverage rises noticeably and reaches its highest levels in the series. This is particularly interesting when compared with the Novelty results. The strong increase in AI-related attention during 2023 was not accompanied by an equally clear increase in average Novelty. The two indicators are therefore describing different changes in the information environment.

Cyber Risk receives a relatively stable but smaller share of news than Climate Risk and, after 2020, Pandemic and Supply Chain Risk. Its Coverage usually remains around 1–2%, although several clear peaks appear over time. Some of the largest increases occur around 2015, 2020, 2021, and 2022. The highest monthly Coverage is observed in early 2022, when the share of Cyber-related news rises above 3%.

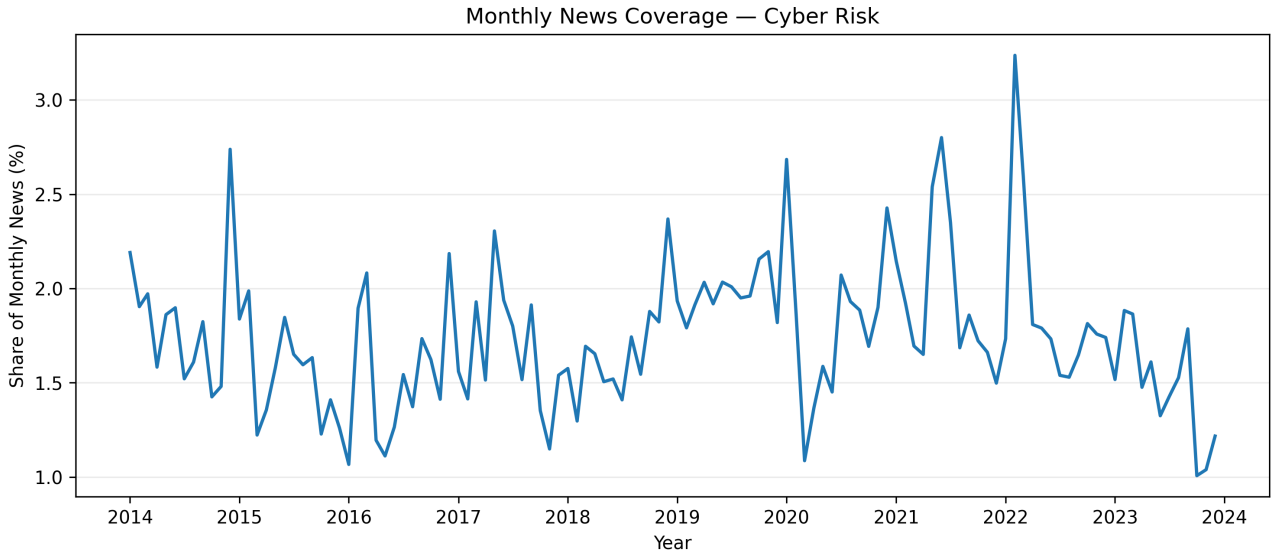


Figure 7.6: Monthly News Coverage for Cyber Risk, 2014–2023

For most of the sample, Cyber Risk accounts for roughly 1–2% of monthly news. This gives the series a relatively stable baseline, although it is interrupted by several short periods of higher attention. An early peak appears around the end of 2014, when Coverage rises well above its usual level. Other increases can be seen during 2016 and 2017, but they are generally temporary.

Cyber Coverage becomes somewhat stronger from 2019 onwards. The series often remains close to or above 2% during this period, and another clear peak appears at the beginning of 2020. Attention then falls sharply before recovering during the rest of the year.
Further increases appear during 2021. The most pronounced change, however, comes at the beginning of 2022. Monthly Coverage rises above 3%, reaching the highest level observed for Cyber Risk in the sample. The increase is short-lived, but it clearly separates this period from the usual level of Cyber-related attention.
After this peak, Coverage gradually returns to lower levels. During 2023, Cyber Risk generally accounts for a smaller share of monthly news, and Coverage falls particularly strongly towards the end of the year. This creates an interesting contrast with the Novelty results. The final months of 2023 were characterised by the highest levels of Cyber Novelty, but not by particularly high News Coverage. In fact, Cyber Coverage was moving in the opposite direction.
The two measures therefore tell different parts of the story. At the beginning of 2022, Cyber Risk receives an unusually high level of news attention. This period is also close to the strongest Semantic Drift identified for Cyber Risk, with the maximum Drift occurring in February 2022. Towards the end of 2023, the situation is different. Cyber-related articles become more novel, but the theme does not receive more attention in the overall news flow. New information can therefore appear even when the risk is not becoming more prominent in the news.

7.4 Burst Attention

While Coverage measures the amount of attention received by each risk, the Burst indicator identifies periods in which this attention increases unusually strongly relative to the risk's own recent history. A burst therefore does not simply correspond to high Coverage. It identifies an increase in attention that is unusually large compared with what was previously normal for that specific risk.
Figure 7.7 shows the evolution of Burst Intensity across the five risk themes. The results show different patterns. Some risks are characterized by one particularly strong episode, while others experience smaller but more recurrent bursts.
Pandemic Risk shows the most extreme burst in the sample. The episode begins in early 2020, when Burst Intensity increases very sharply and reaches almost 70 percentage points at its highest point. It then declines over the following months. This pattern is very different from the other risks. Pandemic Risk is not characterised by several separate bursts, but by one exceptionally large episode concentrated in 2020.

Climate Risk shows a more recurrent pattern. Several bursts appear over the sample, with episodes around 2015, 2017, 2019 and 2021. The strongest occurs in 2017, when Burst Intensity reaches slightly above 5 percentage points. These episodes are much smaller than the Pandemic burst, but they appear at different points in time. Climate Risk therefore experiences repeated periods in which attention increases unusually strongly relative to its own recent history.

Supply Chain Risk also experiences several burst episodes. They appear at different points in the sample and become particularly visible around 2018–2019, 2020 and 2021–2022. The strongest intensity is reached during the 2020 episode, at around 4 percentage points. Some of these episodes also continue for several months, suggesting that the increase in attention can remain unusually high

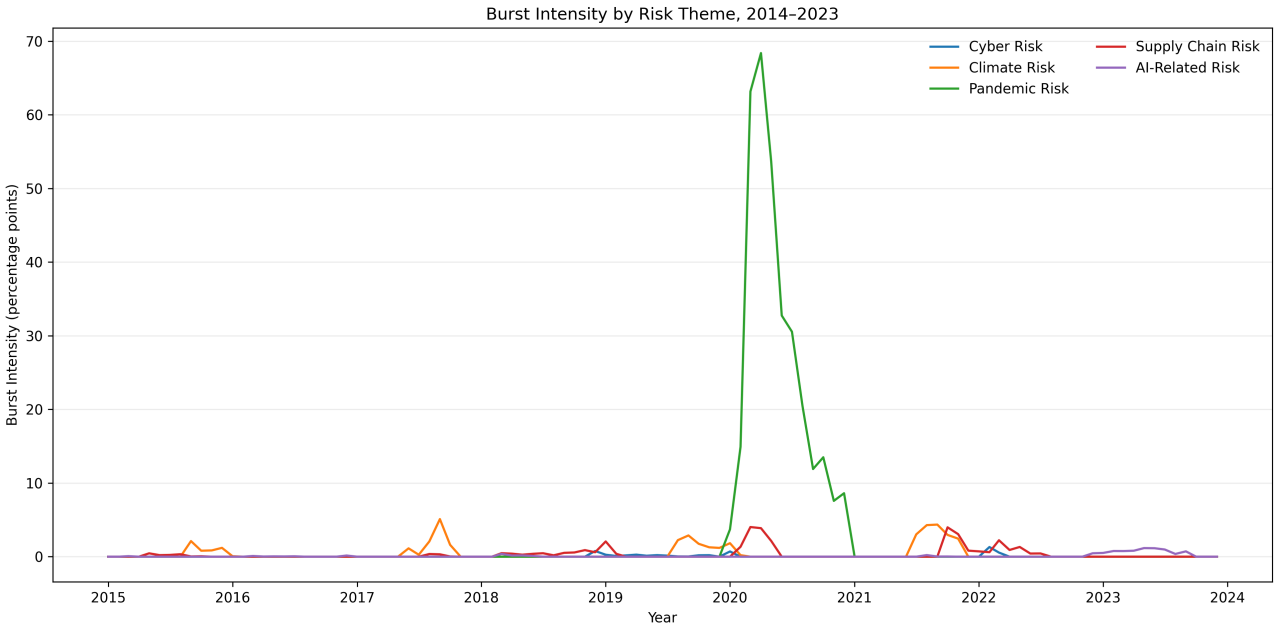


Figure 7.7: Burst Intensity by Risk Theme, 2014–2023

for a relatively extended period.

AI-Related Risk shows bursts that are much smaller in absolute terms. This is consistent with the low level of AI Coverage observed for most of the sample. However, the indicator still detects several separate episodes. The clearest and most persistent one appears between the end of 2022 and 2023, at the same time as the increase in AI Coverage observed in the previous section. The Burst indicator therefore shows that this increase was also unusual relative to the previous history of AI-related attention.

Cyber Risk shows three distinct burst episodes. Their intensity is relatively small compared with the largest episodes observed for Pandemic, Climate and Supply Chain Risk, but they identify specific periods in which Cyber Coverage increased unusually strongly relative to its recent history. The first and longest episode occurs between the end of 2018 and 2019, followed by a short episode at the beginning of 2020. A final burst appears at the beginning of 2022.

Figure 7.8 provides a closer view of these episodes by showing the Cyber Coverage series together with the months classified as part of an active burst.

The first Cyber burst begins in December 2018 and continues until November 2019. It is the longest of the three detected episodes, lasting twelve months. During this period, Coverage remains unusually high relative to its previous history, although it does not increase continuously. This is important for understanding how the indicator works. Once a burst starts, it remains active while Coverage stays above its recent historical mean. A burst can therefore describe a period of sustained abnormal attention, rather than only a single monthly peak.

The second episode occurs in January 2020 and lasts only one month. Coverage reaches approximately 2.7%, but the increase is not sustained in the following months. The episode is therefore much shorter than the one observed in 2019.

The third burst occurs in February and March 2022. Although it lasts only two months, it is the strongest Cyber episode in terms of intensity. Coverage reaches approximately 3.24% in February,

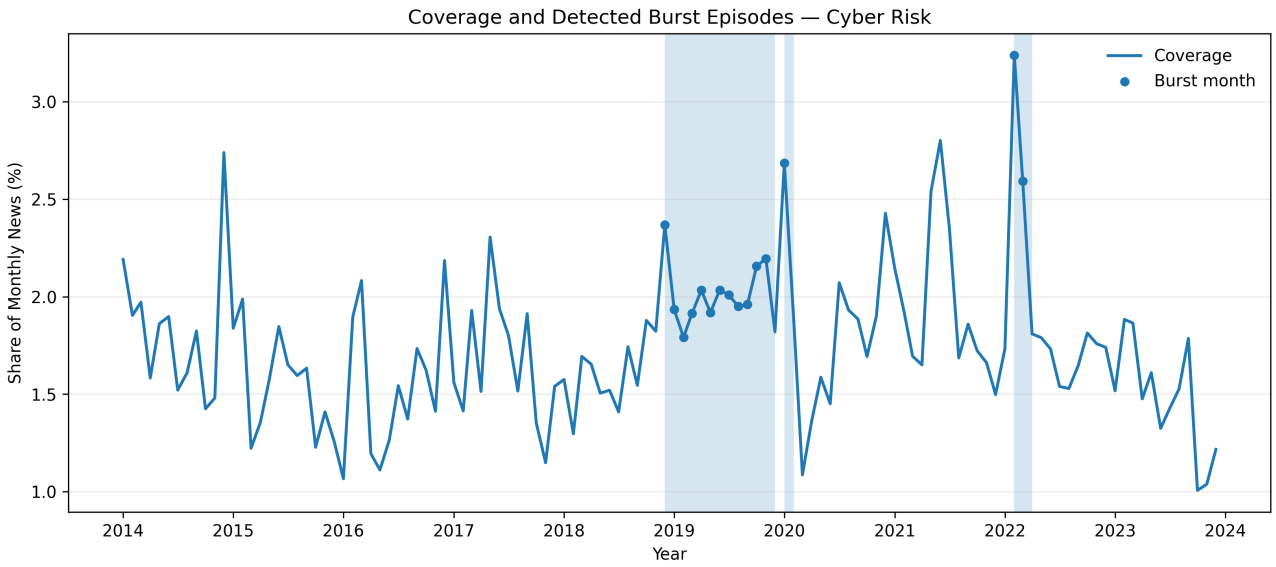


Figure 7.8: News Coverage and Detected Burst Episodes for Cyber Risk, 2014–2023

which is also the highest monthly Coverage observed for Cyber Risk in the sample. Burst Intensity reaches about 1.31 percentage points.

The figure also shows why Burst and Coverage should not be interpreted in the same way. Cyber Coverage contains several visible peaks throughout the sample, but only some of them are classified as bursts. A high level of Coverage is not enough. The increase must be unusual relative to the recent history of Cyber Risk. The indicator therefore separates ordinary changes in attention from periods in which the increase is particularly strong compared with what came before.

The 2022 episode becomes especially interesting when the results are considered together with Semantic Drift. February 2022 is also the month in which Cyber Risk reaches its highest Semantic Drift, at approximately 0.154. The two indicators are measuring different things, but in this case they identify the same period. Attention to Cyber Risk was increasing unusually strongly, while the semantic content associated with the risk was also changing substantially relative to its recent history.

This differs from what is observed towards the end of 2023. The Novelty analysis showed a clear increase in the semantic novelty of Cyber-related articles during the final months of the sample. However, this period is not identified as a Cyber burst and Coverage is relatively low. The different indicators therefore do not always move together. In early 2022, the main signal comes from a change in attention and semantic content, while in late 2023 the strongest signal comes from the arrival of more novel information.